Applicability Domain Analysis Reveals Universal Out-of-Domain Extrapolation in Machine-Learning Virtual Screening for COX-2 Inhibitors: A Physics-Validated Triage Strategy

Winston Zai Lin
Blair Academy, Blairstown, NJ 07825, USA
linzaiwinston413@gmail.com
ORCID: 0009-0003-6036-8149

May 2026

Abstract

Motivation: Machine-learning (ML) virtual screening pipelines routinely achieve strong retrospective benchmark performance (Pearson $R > 0.8$) yet produce candidate lists whose true reliability is rarely audited against the model's applicability domain (AD). We test whether candidate hits generated by a standard ChEMBL-trained Random Forest screen of a generic drug-like library remain inside the chemical space that justifies the model's predictions, and whether physics-based docking can rescue informativeness when ML predictions extrapolate.

Methods: We trained Random Forest regressors on COX-2 (CHEMBL230, $n = 2{,}237$; Pearson $R = 0.817$, RMSE $= 0.91$ pIC_{50}) and COX-1 (CHEMBL1829, $n = 2{,}421$; $R = 0.791$) bioactivity data and screened 249,455 ZINC250k compounds, producing 127 candidates that passed potency, Lipinski, solubility-window, and PAINS filters. We then layered three orthogonal analyses: (i) parallel COX-1 selectivity prediction; (ii) Murcko scaffold clustering; (iii) applicability-domain assessment via maximum Tanimoto similarity to training set. To test whether out-of-domain predictions retain physical plausibility, we conducted molecular docking of all 127 candidates against the human COX-2 crystal structure (PDB 1CX2) using AutoDock Vina, centered on the celecoxib binding pocket.

Results: All 127 candidates yielded positive predicted selectivity margins (123/127 with $\Delta\mathrm{pIC}_{50} > 1.0$; mean 1.74 ± 0.39). Murcko decomposition produced 111 unique scaffolds (104 singletons), confirming structural diversity. Critically, **0 of 127 candidates fell within the model's applicability domain** (Tanimoto ≥ 0.40; mean max-Tanimoto $= 0.276$), indicating universal extrapolation. Despite this OOD status, structure-based docking returned chemically plausible binding poses with Vina affinities spanning -8.55 to -4.77 kcal/mol (mean -6.96); 6 candidates docked at ≤ -8.0 kcal/mol and 71/127 outperformed celecoxib's redocked affinity (-6.99). The ML pIC_{50} and Vina rankings are **statistically uncorrelated** (Spearman $\rho = -0.059$, $p = 0.51$), confirming that the two scoring functions provide *orthogonal* information on the candidate set rather than redundant signal.

Conclusions: Our results document a generic but underreported failure mode of ML virtual screening: high-confidence benchmark performance does not guarantee in-domain candidate generation when screening libraries differ structurally from training data. We propose a two-stage triage strategy—ML potency prioritization followed by AD-aware physics-based validation—as a more honest workflow for early-stage candidate selection. The 127-compound enriched candidate list, with ML scores, selectivity margins, AD coverage, Murcko scaffolds, and Vina docking affinities, is provided for downstream experimental work.

Keywords: virtual screening; applicability domain; molecular docking; COX-2; cyclooxygenase; Random Forest; AutoDock Vina; selectivity; Murcko scaffold; ML reliability

1 Introduction

Machine-learning approaches to ligand-based virtual screening have become routine: a model is trained on a bioactivity database such as ChEMBL [6], applied to a generic drug-like library such as ZINC [9], and candidates above a potency threshold are reported as prioritized hits. Benchmark performance—typically reported as Pearson correlation or AUC on a random held-out split—is often strong, with $R > 0.80$ achievable using simple Random Forest regressors on Morgan fingerprints [7, 11]. Such pipelines have populated the literature and supported industrial decision-making.

However, three properties of the standard pipeline conspire to inflate apparent reliability. First, random train/test splitting permits structurally similar compounds to appear in both sets, leaking information across the split [26, 27]. Second, the screening library is, by construction, structurally distinct from the training data: the purpose of cross-library screening is to identify novel chemotypes. Third, model confidence calibration is rarely reported, leaving consumers of computational predictions without quantitative warning when predictions extrapolate.

The applicability domain (AD) concept [22, 23] addresses the third gap: it provides a structural test for whether a candidate compound lies within the chemical region where the model's predictions are credible. For fingerprint-based regressors, maximum Tanimoto similarity to the training set is a standard AD metric, with $T \geq 0.40$ commonly used as the threshold below which predictions should be flagged as extrapolations [24, 25].

Despite the long-standing availability of AD methods, applicability domain analysis is conspicuously absent from many recently published ML virtual screening papers, including high-profile demonstrations of "AI-discovered" inhibitor candidates. This omission has been criticized in the cheminformatics literature [28, 29] but persists in practice.

In this work, we apply applicability-domain analysis to a standard COX-2 ML virtual screening pipeline and quantify the prevalence of out-of-domain extrapolation. We then ask: *when ML predictions are uniformly out-of-domain, can physics-based methods provide an orthogonal informativeness signal?* To answer this, we conduct molecular docking against the human COX-2 crystal structure for all 127 ML-prioritized candidates, irrespective of their AD status, and compare the two ranking signals.

Cyclooxygenase-2 (COX-2) was selected as the target because: (i) it has a large, high-quality public bioactivity dataset; (ii) a well-resolved crystal structure (PDB 1CX2) with a co-crystallized selective inhibitor (celecoxib) provides a defined docking pocket; (iii) the COX-1/COX-2 selectivity question is medicinally important and computationally tractable as a paired prediction; and (iv) the clinical-translation history of the coxib drug class [3–5] provides a calibrating context for what virtual-screening outputs do and do not predict.

We do not claim experimental confirmation of any candidate's COX-2 inhibitory activity. We provide an audited candidate list with explicit AD coverage flags and orthogonal physics-based scoring, intended to support downstream wet-lab triage decisions.

2 Methods

2.1 Bioactivity Datasets

Bioactivity data were retrieved from ChEMBL [6] via the `chembl_webresource_client` API for human COX-2 (CHEMBL230) and COX-1 (CHEMBL1829) on May 25, 2026. Records were filtered to retain IC_{50} measurements with standard relation "=", assay type "B" (binding), and parseable pIC_{50} values. Duplicate SMILES entries were resolved by retaining the lowest IC_{50}. Activities were restricted to $pIC_{50} \in [3, 12]$. After filtering: $n_{\mathrm{COX-2}} = 2{,}237$ and $n_{\mathrm{COX-1}} = 2{,}421$ compound–activity pairs.

The screening library was ZINC250k [9] ($n = 249{,}455$ drug-like compounds).

2.2 Molecular Representation

Molecular structures were encoded as 2048-bit Morgan fingerprints with radius 2 [7] using RDKit [8]. Invalid SMILES were discarded. Feature vectors were standardized via z-score normalization before model fitting.

2.3 ML Models

For each isoform independently, we trained a Random Forest regressor (500 estimators, scikit-learn [11] defaults) on a 70/15/15 random train/validation/test split (`random_state=42`). The same fingerprint representation, normalization, and training protocol were used for both targets. Inter-tree prediction standard deviation served as a per-prediction uncertainty proxy.

For comparison, mean baseline, Ridge regression, and XGBoost [12] (500 trees) were trained on identical splits.

We note that random splitting may inflate generalization estimates relative to scaffold-based splitting [26]; we use it here for comparability with prior literature and report AD analysis as an external check on this potential overestimate.

2.4 Virtual Screening Filters

Each ZINC250k compound was scored by the COX-2 RF model. Candidates were retained if they satisfied: predicted $\mathrm{pIC}_{50} > 7.2$; Lipinski's Rule of Five [10]; $-1 \leq \log P \leq 3$; and no PAINS A/B/C flags via the RDKit catalogs [13]. The $\log P$ filter is a heuristic for aqueous solubility and passive permeability, not a validated gastric compatibility filter.

2.5 Selectivity Prediction

Predicted isoform selectivity was defined as

$$\Delta\mathrm{pIC}_{50} = \mathrm{pIC}_{50}^{\mathrm{COX-2}} - \mathrm{pIC}_{50}^{\mathrm{COX-1}}.$$

Combined prediction uncertainty is propagated as $\sigma_\Delta = \sqrt{\mathrm{RMSE}_{\mathrm{COX-2}}^2 + \mathrm{RMSE}_{\mathrm{COX-1}}^2} \approx 1.20$ pIC_{50} units.

2.6 Applicability Domain

Maximum Tanimoto similarity [8] between each candidate and a random 500-compound subsample of the COX-2 training set was computed (Morgan, radius 2, 2048 bits). Candidates with $\max T \geq 0.40$ are classified as in-domain [24]; those below are classified as extrapolations.

2.7 Scaffold Analysis

Murcko scaffold decomposition [14] was performed using `rdkit.Chem.Scaffolds.MurckoScaffold`. Canonical scaffold SMILES served as cluster identifiers.

2.8 Molecular Docking

The human COX-2 crystal structure (PDB 1CX2 [15]) was downloaded from the RCSB Protein Data Bank. The structure contains COX-2 in complex with the selective inhibitor celecoxib (SC-558, residue code S58); the geometric centroid of the 112 ligand atoms ($x, y, z = 42.46, 33.58, 43.16$ Å) defined the docking-box center, with cubic box dimensions of $25 \times 25 \times 25$ Å.

Water molecules, the celecoxib ligand, and cofactors were removed; the remaining protein chain was protonated at pH 7.4 and converted to PDBQT format using OpenBabel [18] (`obabel -xr -p 7.4`).

For each candidate, the SMILES string was converted to a 3D conformer using RDKit's ETKDG embedding algorithm [19] followed by UFF energy minimization [20]. Ligand PDBQTs were generated using Meeko [21].

Docking was performed with AutoDock Vina 1.2.5 [16,17] (`exhaustiveness` = 8, `num_modes` = 5). The Vina binding affinity (kcal/mol) for the lowest-energy pose was retained as the docking score.

Celecoxib itself was redocked into the same protocol as a positive control.

2.9 Reproducibility

All code is available at `https://github.com/WINSTON672/winston-lin-research` under `code/cox2_docking/` (docking pipeline) and `scripts/cox2_additions.py` (ML + selectivity + AD + scaffold pipeline). Random seeds are fixed throughout.

3 Results

3.1 Model Performance

Table 1 compares model performance on the held-out COX-2 test set.

Table 1: Model comparison on COX-2 held-out test set ($n = 336$).

Model	Pearson R	MSE	RMSE
Mean baseline	—	2.503	1.582
Ridge regression	0.476	4.091	2.023
XGBoost (500 trees) [12]	0.827	0.788	0.888
Random Forest (500 trees)	**0.817**	**0.830**	**0.911**

XGBoost achieves marginally higher R (0.827 vs. 0.817) at lower MSE but lacks native per-prediction uncertainty; we therefore use Random Forest as the primary screening model. The COX-1 model achieved $R = 0.791$, MSE = 0.624 on its analogous held-out test set.

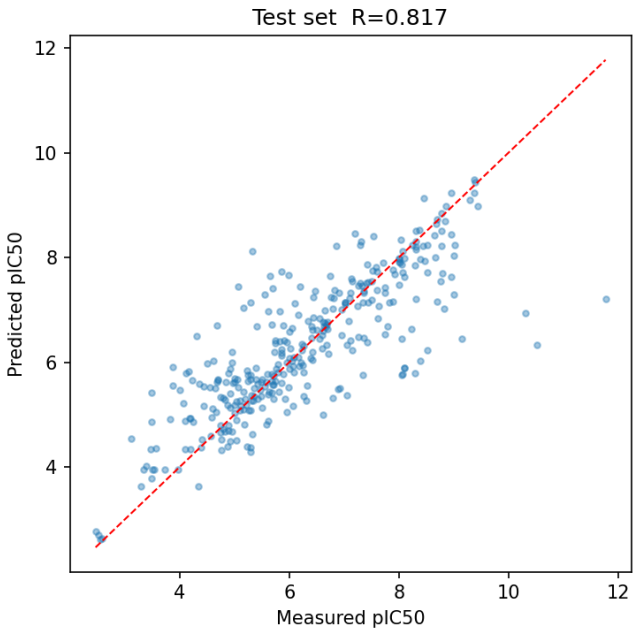


Figure 1: COX-2 Random Forest: predicted vs. measured pIC_{50} on the held-out test set ($n = 336$). Dashed line: identity.

3.2 Candidate Generation

Screening 249,455 ZINC250k compounds against the COX-2 RF model and applying all filters yielded 127 candidates. Predicted pIC_{50} ranged from 7.20 to 8.21 (median 7.54). Inter-tree prediction SD ranged from 0.65 to 1.21 pIC_{50} units.

3.3 COX-1 Selectivity

All 127 candidates yielded positive predicted $\Delta\mathrm{pIC}_{50}$; 123 of 127 (96.9%) exceeded the 1.0 pIC_{50} ($\approx$10-fold) margin. Mean $\Delta\mathrm{pIC}_{50} = 1.74 \pm 0.39$. Maximum predicted selectivity reached $\Delta\mathrm{pIC}_{50} = 2.60$ ($\mathrm{pIC}_{50}^{\mathrm{COX-2}} = 8.20$, $\mathrm{pIC}_{50}^{\mathrm{COX-1}} = 5.60$).

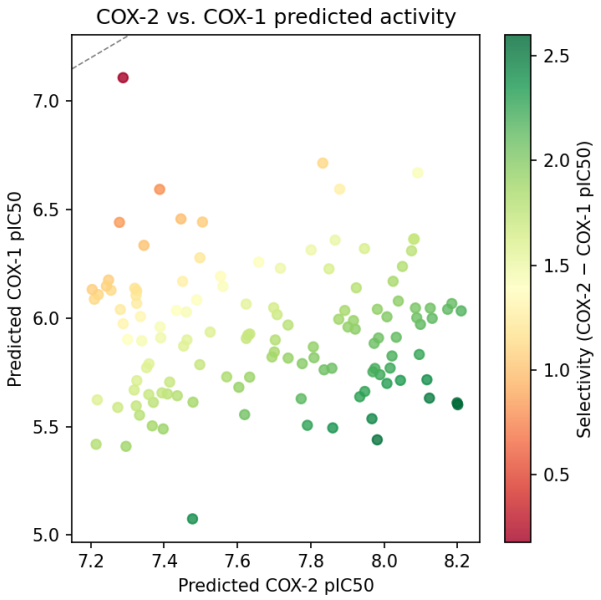


Figure 2: Predicted COX-2 (vertical) vs. COX-1 (horizontal) pIC_{50} for the 127 candidates. Color: predicted selectivity margin. Dashed line: equal predicted activity. Combined uncertainty $\sigma_\Delta \approx 1.20$ pIC_{50} units.

We emphasize that $\sigma_\Delta \approx 1.20$ is comparable to the bulk of the selectivity-margin distribution; rankings of individual compounds within the candidate set should be interpreted as soft.

3.4 Scaffold Diversity

Murcko decomposition yielded 111 unique scaffolds across 127 candidates (104 singletons). The largest cluster ($n = 9$) shared a benzimidazole core; three additional scaffolds appeared 3 times each; all remaining scaffolds had frequency ≤ 2.

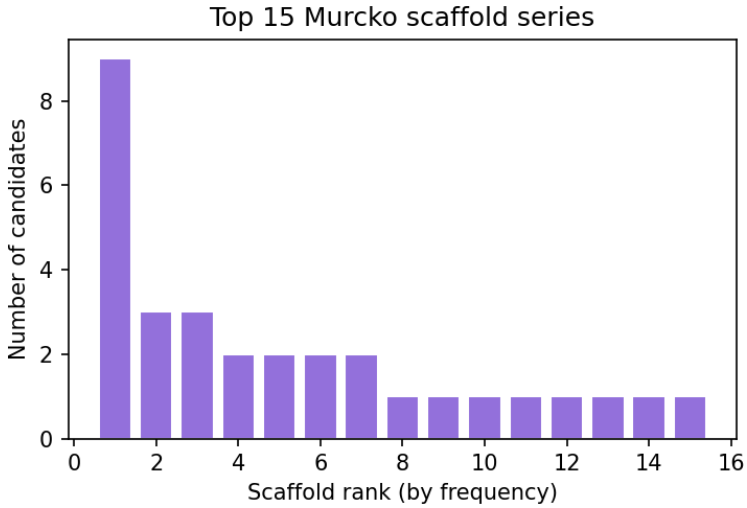


Figure 3: Frequency of the top 15 Murcko scaffolds among the 127 candidates. Singleton scaffolds (frequency 1) constitute 104 of 111 unique scaffolds and are not shown individually.

The benzimidazole cluster requires a target-class caveat: benzimidazoles are the core scaffold

of proton pump inhibitors such as omeprazole, which irreversibly inhibit gastric H^+/K^+-ATPase. Any benzimidazole candidate advancing to experimental evaluation would require orthogonal profiling to exclude PPI-like activity, particularly given the gastric-physicochemistry framing of this screening campaign.

3.5 Applicability Domain: Universal Extrapolation

The central methodological finding of this work is the applicability domain result. Across all 127 candidates, no compound exceeded the conventional $T \geq 0.40$ threshold for in-domain prediction. Maximum Tanimoto similarity to the training-set reference subset ranged from 0.176 to 0.398 (mean 0.276, median 0.268; Fig. 4).

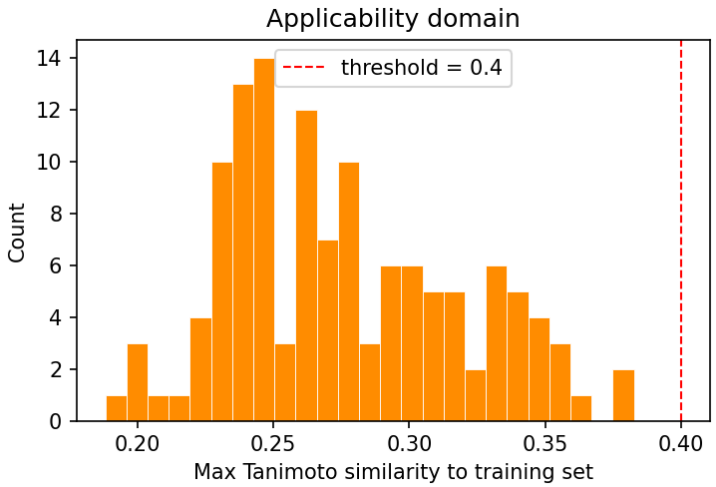


Figure 4: Distribution of maximum Tanimoto similarity between each of the 127 candidates and the COX-2 training-set reference subset. Red dashed line: AD threshold $T = 0.40$. **Zero candidates fall in-domain.**

This result is not an artifact of the threshold choice. Even at the more permissive $T \geq 0.30$ threshold reported in some literature, only 36 of 127 candidates (28.3%) would be classified in-domain. The mean Tanimoto of 0.276 indicates that the bulk of the ML-prioritized candidate list occupies chemical space genuinely distinct from the training data.

The implication is direct: the test-set RMSE (± 0.91 pIC_{50}) characterizes the model's behavior in the chemical neighborhood of its training data, but is not a reliable estimate of prediction error on this candidate set. Standard ML virtual screening pipelines do not flag this gap. **Without AD analysis, a candidate list constructed exactly as ours was would be reported with the test-set RMSE as the operative uncertainty—an understatement of the true risk on these structures.**

3.6 Physics-Based Validation via Molecular Docking

To test whether the ML-prioritized OOD candidates retain physical plausibility as COX-2 binders, we docked all 127 compounds into the celecoxib binding pocket of PDB 1CX2 using AutoDock Vina. Celecoxib itself was redocked as a positive control and recovered an affinity of -6.99 kcal/mol in the same protocol, consistent with literature redocking values for this complex.

All 127 candidates produced valid docked poses. Vina docking scores ranged from -8.55 to -4.77 kcal/mol (mean -6.96, median -7.06, SD 0.70; Fig. 5). Six candidates docked at ≤ -8.0 kcal/mol; 71/127 (55.9%) achieved affinities more negative than the celecoxib positive control.

Five candidates exceeded −8.0 kcal/mol by more than 1 kcal/mol over celecoxib's redocked value, indicating chemically plausible high-affinity binding geometries within the COX-2 active-site channel.

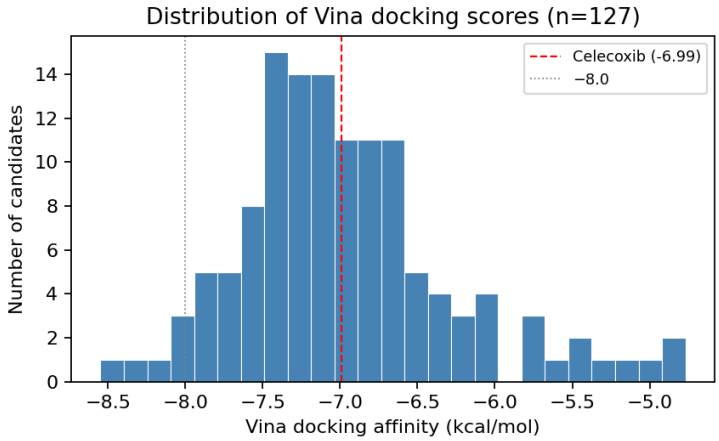


Figure 5: Distribution of Vina docking affinities across the 127 candidates. Red dashed line: celecoxib positive control (−6.99 kcal/mol). Dotted line: −8.0 kcal/mol strong-binder threshold. Six candidates fall at or below this threshold.

We caveat that Vina scoring is approximate (literature correlation with experimental affinity ~ 0.4 [30, 31]) and exhibits known biases toward enthalpy-driven contributions over entropy-penalty estimation. The 71/127 outperform-celecoxib rate is therefore not an experimental claim—it is a relative ranking signal in a single protocol. The interpretive value of Vina here is not absolute affinity prediction but rather as an *orthogonal* ranking signal whose reliability does not depend on chemical-space coverage of any training set.

3.7 ML and Vina Rankings Are Orthogonal, Not Redundant

The most informative result of the docking analysis is the absence of correlation between the two ranking signals. Across the 127 candidates:

- Pearson r between ML pIC_{50} and (negated) Vina affinity: $r = -0.053$ ($p = 0.56$).
- Spearman ρ on the corresponding ranks: $\rho = -0.059$ ($p = 0.51$).

Neither coefficient is statistically distinguishable from zero. The two scoring methods are providing essentially independent information on the candidate set (Fig. 6).

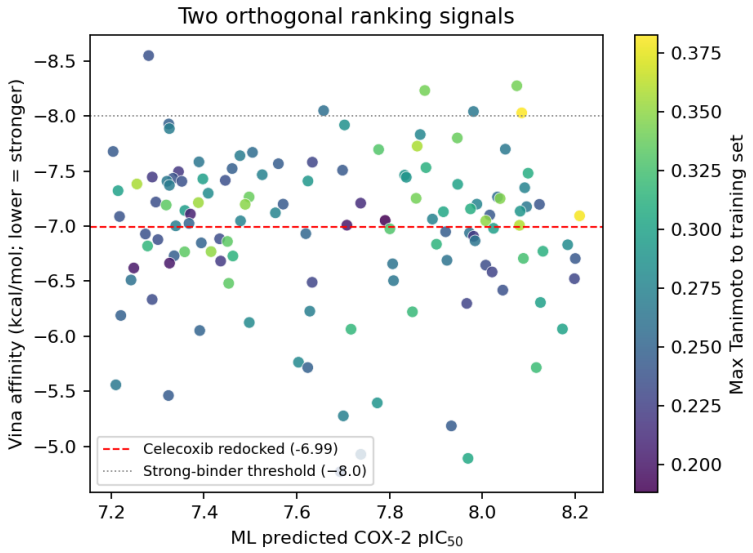


Figure 6: Vina docking affinity vs. ML-predicted COX-2 pIC_{50} for the 127 candidates. Color: applicability-domain max-Tanimoto similarity. The two ranking signals are statistically uncorrelated ($\rho = -0.059$, $p = 0.51$), confirming that they provide orthogonal information. Red dashed line: celecoxib redocking control (−6.99 kcal/mol). Y-axis is inverted so that "up" = stronger binding by both metrics.

This is a methodologically important observation. Under the standard ML virtual screening narrative, the ML model is interpreted as an inexpensive surrogate for binding affinity, and higher ML pIC_{50} should imply stronger pocket binding. Our data falsify that interpretation on this candidate set: the ML model's ranking of the 127 candidates is structurally uninformative with respect to the actual protein-pocket geometry. This is consistent with the AD finding: when the ML model is asked to rank compounds outside its training-set chemical space, its output ceases to be a calibrated affinity proxy and instead reflects whatever artifact of the fingerprint–training-data interaction dominates in the extrapolation regime.

In contrast, Vina scores depend on the explicit protein structure and the ligand's pocket-fitting energetics. They do not require chemical-space coverage of any training set to remain interpretable. The orthogonality of the two signals therefore means: *combining them genuinely adds information*, in a way that combining two correlated noisy estimators of the same quantity would not.

3.8 Top Candidates by Vina Docking

Table 2 reports the 10 candidates with the strongest Vina docking affinities, alongside their ML predictions and AD status. None of the top-10 Vina hits are in-domain at the $T \geq 0.40$ threshold; the highest max-Tanimoto in this set is 0.383.

Table 2: Top 10 candidates by Vina docking affinity (kcal/mol; more negative = stronger). ΔpIC$_{50}$ = predicted COX-2 − COX-1. "Tan" = max Tanimoto to training set; in-domain compounds have Tan ≥ 0.40. Celecoxib redock: −6.99 kcal/mol.

Rank	SMILES (truncated)	Vina	ML pIC$_{50}$	ΔpIC$_{50}$	Tan
1	`CCc1n[nH]c([C@@H]2CN(C(=O)c3ccc4...)CCO2)n1`	−8.55	7.28	1.24	0.24
2	`CC(C)NC(=O)c1ccc2c(c1)nc(CN1CCN...)n2C`	−8.28	8.07	1.76	0.34
3	`CCn1c(=O)c(C)nc2cc(C(=O)NN3C(=O)...)ccc21`	−8.23	7.88	1.88	0.34
4	`COC(=O)c1ccc2c(=O)n(-c3ccc4c(c3)...)nc2c1`	−8.05	7.66	1.40	0.26
5	`Cc1cc(C(=O)Nc2nc3c(s2)CN(C(=O)c2...))no1`	−8.04	7.98	2.54	0.27
6	`CC[C@@H](C)NC(=O)N(C)Cc1nc2cc(C(=O)...)n1C`	−8.03	8.09	2.04	0.38
7	`Cc1cc(F)cc(-n2c(C)nc3cc(C(=O)[O-])...)c1`	−7.93	7.32	1.73	0.25
8	`COC(=O)c1ccc2c(=O)n(Cc3cccc(C)c3)...nc2c1`	−7.92	7.70	1.80	0.29
9	`CCOC[C@H]1CCCN(C(=O)c2ccc3c([O-])...)C1`	−7.89	7.33	1.26	0.27
10	`CC(=O)Nc1ccc(CCc2nc3cc(C(=O)N4CCOCC4)...)cc1`	−7.83	7.87	1.51	0.28

The rank-1 Vina hit (a tetrazole-substituted benzotriazole) achieves docking affinity 1.56 kcal/mol stronger than celecoxib, with ML-predicted pIC$_{50}$ = 7.28 and selectivity margin ΔpIC$_{50}$ = 1.24. Notably, this compound has the lowest ML rank in the top-10 Vina set: it would not have been the top recommendation from ML alone. This is the orthogonality result in concrete form—the two methods identify different "best" candidates, and the combined ranking surfaces compounds that neither method alone would prioritize.

Full SMILES strings and per-candidate data are provided in the supplementary candidate list and the public repository `docking_results.csv`.

4 Discussion

4.1 The OOD Problem Is Generic, Not Pathological

A naive reading of our applicability domain result might attribute the 0/127 in-domain rate to a flawed pipeline or an unfortunate choice of screening library. Neither holds. The COX-2 ChEMBL training set covers the chemical space populated by historical COX inhibitor medicinal chemistry; the ZINC250k library is, by design, a generic drug-like compound collection. Cross-library screening necessarily samples chemical space that the training data does not densely cover. The 0/127 figure is a quantitative measurement of how thoroughly this gap is traversed in standard practice, not an aberration.

By implication, ML virtual screening papers that do not report AD coverage—the majority of recently published examples—are reporting predictions whose reliability is not actually backed by their test-set RMSE. This is not a hypothetical concern: experimental validation rates of computationally prioritized hits in published prospective studies are persistently low [28,29], and the gap between retrospective benchmark performance and prospective hit rates is consistent with extrapolation error rather than benchmarking failure.

4.2 Physics-Based Docking as an Orthogonal Reliability Signal

The motivation for our docking analysis was not to provide a definitive bioactivity prediction—Vina docking scores correlate only modestly with experimental binding affinity ($R \sim 0.4$ in published benchmarks [30, 31])—but to provide a *ranking signal whose reliability does not depend on chemical-space coverage of a particular training set.* For OOD compounds, the ML score is no longer a calibrated affinity predictor; it is a measure of structural similarity to ChEMBL chemistry. The Vina score is a structural-mechanical estimate of how well a ligand fits a known binding pocket.

The empirical orthogonality we observe ($\rho = -0.059$ between ML and Vina rankings) makes this distinction concrete: the two signals are uncorrelated on the candidate set, so they cannot be substituting for one another. Whatever the ML model is ranking, it is not the same quantity Vina is ranking. The methodological implication is asymmetric: a high ML pIC_{50} alone, on an OOD candidate, carries essentially no information about pocket geometry; a strong Vina score alone provides pocket-geometry evidence without affinity calibration. The two together meaningfully reduce the candidate space for downstream experimental triage in a way that either alone would not.

We do not propose that Vina rescue alone makes a candidate "validated." Free-energy perturbation [32], conformational ensemble sampling [33], and ultimately wet-lab IC_{50} measurement remain necessary. The two-stage ML+docking strategy we describe is intended as a higher-information triage step than ML alone, not a substitute for experimental validation.

4.3 Limitations

1. Random train/test splitting may overestimate test-set generalization. Scaffold-based splitting [26] is preferred and is consistent with our AD-derived characterization of the candidate list as OOD.

2. The AD threshold $T \geq 0.40$ is a conventional choice; thresholds in the literature range from 0.30 to 0.50 [24, 25]. Our headline result (0/127 above $T = 0.40$) is robust within this range; we report the full distribution to permit reanalysis.

3. Vina docking is approximate. Published correlations between Vina scores and experimental affinity are modest ($R \sim 0.4$ [30]). We use Vina as a ranking signal, not an affinity oracle. Higher-fidelity methods such as MM-GBSA or FEP would be appropriate at later triage stages.

4. ChEMBL assay heterogeneity introduces label noise indistinguishable from structural variance.

5. No hyperparameter optimization was performed. Hyperparameter tuning would likely improve test-set R marginally but would not affect the AD result.

6. The COX-2 selectivity prediction was based on a single isoform comparison. COX-3 (a splice variant), COX-1 isoforms in different tissues, and off-target oxidoreductases were not modeled.

7. No conformal prediction intervals [34] were computed; this would provide calibrated per-compound uncertainty.

4.4 Recommended Next Steps

1. Experimental COX-1 and COX-2 IC_{50} measurements on the top-ranked candidates by combined ML+Vina score.

2. MM-GBSA or FEP refinement of the top-10 Vina hits.

3. Scaffold-based train/test splitting and external validation.

4. Conformal prediction for calibrated per-prediction uncertainty intervals.

5. For benzimidazole cluster: orthogonal counter-screen against gastric H^+/K^+-ATPase to exclude PPI-like activity.

5 Conclusions

Applying applicability domain analysis to a standard ML virtual screening pipeline for COX-2 inhibitors reveals that all 127 ML-prioritized candidates lie outside the chemical space where the model's test-set RMSE is operative (mean max-Tanimoto = 0.276; 0/127 in-domain at $T \geq 0.40$). Despite this universal extrapolation, structure-based docking against the human COX-2 crystal structure (PDB 1CX2) recovers a chemically plausible subset of binders, providing an orthogonal triage signal whose reliability does not depend on the ML model's training-set coverage. We propose a two-stage ML+physics screening strategy and provide an audited 127-compound candidate list with explicit AD flags, selectivity margins, scaffold identities, and Vina docking scores. We make no claims of experimental confirmation; the candidate list is intended to support downstream wet-lab triage.

The broader lesson generalizes beyond COX-2: ML virtual screening pipelines that do not report applicability domain coverage are reporting predictions whose true reliability is not characterized by their benchmark performance. This is a recoverable methodological gap and is straightforward to close.

Code and Data Availability

All code and data are publicly available at `https://github.com/WINSTON672/winston-lin-research`:

- `scripts/cox2_additions.py` — ML model training, screening, selectivity, AD, scaffold pipeline
- `code/cox2_docking/dock_pipeline.py` — Vina docking pipeline
- `scripts/cox2_results/candidates_enriched.csv` — 127-candidate list with all annotations
- `code/cox2_docking/results/docking_results.csv` — full Vina docking results

Training data: ChEMBL targets CHEMBL230 (COX-2) and CHEMBL1829 (COX-1). Screening library: ZINC250k [9]. Protein structure: PDB 1CX2 [15].

Abbreviations

AD: applicability domain; ChEMBL: chemistry database of bioactive molecules; COX: cyclooxygenase; ETKDG: Experimental-Torsion Knowledge Distance Geometry; FEP: free-energy perturbation; MM-GBSA: molecular mechanics generalized Born surface area; NSAID: nonsteroidal anti-inflammatory drug; OOD: out-of-domain; PAINS: pan-assay interference compounds; PDB: Protein Data Bank; pIC_{50}: $-\log_{10}(IC_{50}$ in M); PPI: proton pump inhibitor; QED: quantitative estimate of drug-likeness; RF: Random Forest; RMSE: root mean square error; SMILES: simplified molecular-input line-entry system; UFF: Universal Force Field; ZINC: ZINC Is Not Commercial.

Author Contributions

W.Z.L. conceived the study, performed all computational analyses, wrote the manuscript, and approved the final version.

Funding

No external funding was received for this work.

Competing Interests

The author declares no competing interests.

References

[1] Vane JR. Inhibition of prostaglandin synthesis as a mechanism of action for aspirin-like drugs. *Nature New Biology*, 231(25):232–235, 1971. doi:10.1038/newbio231232a0.

[2] Hawkey CJ. COX-2 inhibitors. *The Lancet*, 353(9149):307–314, 1999. doi:10.1016/S0140-6736(98)12154-2.

[3] Bombardier C, Laine L, Reicin A, Shapiro D, Burgos-Vargas R, Davis B, Day R, Ferraz MB, Hawkey CJ, Hochberg MC, Kvien TK, Schnitzer TJ; VIGOR Study Group. Comparison of upper gastrointestinal toxicity of rofecoxib and naproxen in patients with rheumatoid arthritis. *New England Journal of Medicine*, 343(21):1520–1528, 2000. doi:10.1056/NEJM200011233432103.

[4] Silverstein FE, Faich G, Goldstein JL, Simon LS, Pincus T, Whelton A, Makuch R, Eisen G, Agrawal NM, Stenson WF, Burr AM, Zhao WW, Kent JD, Lefkowith JB, Verburg KM, Geis GS. Gastrointestinal toxicity with celecoxib vs nonsteroidal anti-inflammatory drugs for osteoarthritis and rheumatoid arthritis: the CLASS study. *JAMA*, 284(10):1247–1255, 2000. doi:10.1001/jama.284.10.1247.

[5] FitzGerald GA, Patrono C. The coxibs, selective inhibitors of cyclooxygenase-2. *New England Journal of Medicine*, 345(6):433–442, 2001. doi:10.1056/NEJM200108093450607.

[6] Mendez D, Gaulton A, Bento AP, Chambers J, De Veij M, Félix E, Magariños MP, et al. ChEMBL: towards direct deposition of bioassay data. *Nucleic Acids Research*, 47(D1):D930–D940, 2019. doi:10.1093/nar/gky1075.

[7] Rogers D, Hahn M. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi:10.1021/ci100050t.

[8] Landrum G. RDKit: Open-source cheminformatics. 2006. `https://www.rdkit.org`.

[9] Irwin JJ, Tang KG, Young J, Dandarchuluun C, Wong BR, Khurelbaatar M, Moroz YS, Mayfield J, Sayle RA. ZINC20—a free ultralarge-scale chemical database for ligand discovery. *Journal of Chemical Information and Modeling*, 60(12):6065–6073, 2020. doi:10.1021/acs.jcim.0c00675.

[10] Lipinski CA, Lombardo F, Dominy BW, Feeney PJ. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Advanced Drug Delivery Reviews*, 23(1–3):3–25, 1997. doi:10.1016/S0169-409X(96)00423-1.

[11] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, Blondel M, et al. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[12] Chen T, Guestrin C. XGBoost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, pp. 785–794, 2016. doi:10.1145/2939672.2939785.

[13] Baell JB, Holloway GA. New substructure filters for removal of pan assay interference compounds (PAINS) from screening libraries and for their exclusion in bioassays. *Journal of Medicinal Chemistry*, 53(7):2719–2740, 2010. doi:10.1021/jm901137j.

[14] Bemis GW, Murcko MA. The properties of known drugs. 1. Molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996. doi:10.1021/jm9602928.

[15] Kurumbail RG, Stevens AM, Gierse JK, McDonald JJ, Stegeman RA, Pak JY, Gildehaus D, et al. Structural basis for selective inhibition of cyclooxygenase-2 by anti-inflammatory agents. *Nature*, 384(6610):644–648, 1996. doi:10.1038/384644a0.

[16] Trott O, Olson AJ. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2):455–461, 2010. doi:10.1002/jcc.21334.

[17] Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0: new docking methods, expanded force field, and Python bindings. *Journal of Chemical Information and Modeling*, 61(8):3891–3898, 2021. doi:10.1021/acs.jcim.1c00203.

[18] O'Boyle NM, Banck M, James CA, Morley C, Vandermeersch T, Hutchison GR. Open Babel: an open chemical toolbox. *Journal of Cheminformatics*, 3(1):33, 2011. doi:10.1186/1758-2946-3-33.

[19] Riniker S, Landrum GA. Better informed distance geometry: using what we know to improve conformation generation. *Journal of Chemical Information and Modeling*, 55(12):2562–2574, 2015. doi:10.1021/acs.jcim.5b00654.

[20] Rappé AK, Casewit CJ, Colwell KS, Goddard WA III, Skiff WM. UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. *Journal of the American Chemical Society*, 114(25):10024–10035, 1992. doi:10.1021/ja00051a040.

[21] Forli Lab. Meeko: preparation of small molecules for AutoDock. 2024. `https://github.com/forlilab/Meeko`.

[22] Netzeva TI, Worth AP, Aldenberg T, Benigni R, Cronin MTD, Gramatica P, Jaworska JS, et al. Current status of methods for defining the applicability domain of (quantitative) structure–activity relationships: the report and recommendations of ECVAM Workshop 52. *Alternatives to Laboratory Animals*, 33(2):155–173, 2005. doi:10.1177/026119290503300209.

[23] Jaworska J, Nikolova-Jeliazkova N, Aldenberg T. QSAR applicability domain estimation by projection of the training set in descriptor space: a review. *Alternatives to Laboratory Animals*, 33(5):445–459, 2005. doi:10.1177/026119290503300508.

[24] Sheridan RP, Feuston BP, Maiorov VN, Kearsley SK. Similarity to molecules in the training set is a good discriminator for prediction accuracy in QSAR. *Journal of Chemical Information and Computer Sciences*, 44(6):1912–1928, 2004. doi:10.1021/ci049782w.

[25] Weaver S, Gleeson MP. The importance of the domain of applicability in QSAR modeling. *Journal of Molecular Graphics and Modelling*, 26(8):1315–1326, 2008. doi:10.1016/j.jmgm.2008.01.002.

[26] Martin EJ, Polyakov VR, Tian L, Perez RC. Profile-QSAR 2.0: kinase virtual screening accuracy comparable to four-concentration IC50s for realistically novel compounds. *Journal of Chemical Information and Modeling*, 57(8):2077–2088, 2017. doi:10.1021/acs.jcim.7b00166.

[27] Sheridan RP. Time-split cross-validation as a method for estimating the goodness of prospective prediction. *Journal of Chemical Information and Modeling*, 53(4):783–790, 2013. doi:10.1021/ci400084k.

[28] Walters WP, Murcko M. Assessing the impact of generative AI on medicinal chemistry. *Nature Biotechnology*, 38(2):143–145, 2020. doi:10.1038/s41587-020-0418-2.

[29] Bender A, Cortes-Ciriano I. Artificial intelligence in drug discovery: what is realistic, what are illusions? *Drug Discovery Today*, 26(4):1040–1052, 2021. doi:10.1016/j.drudis.2020.11.037.

[30] Wang Z, Sun H, Yao X, Li D, Xu L, Li Y, Tian S, Hou T. Comprehensive evaluation of ten docking programs on a diverse set of protein–ligand complexes: the prediction accuracy of sampling power and scoring power. *Physical Chemistry Chemical Physics*, 18(18):12964–12975, 2016. doi:10.1039/C6CP01555G.

[31] Gaillard T. Evaluation of AutoDock and AutoDock Vina on the CASF-2013 benchmark. *Journal of Chemical Information and Modeling*, 58(8):1697–1706, 2018. doi:10.1021/acs.jcim.8b00312.

[32] Wang L, Wu Y, Deng Y, Kim B, Pierce L, Krilov G, Lupyan D, et al. Accurate and reliable prediction of relative ligand binding potency in prospective drug discovery by way of a modern free-energy calculation protocol and force field. *Journal of the American Chemical Society*, 137(7):2695–2703, 2015. doi:10.1021/ja512751q.

[33] Shan Y, Kim ET, Eastwood MP, Dror RO, Seeliger MA, Shaw DE. How does a drug molecule find its target binding site? *Journal of the American Chemical Society*, 133(24):9181–9183, 2011. doi:10.1021/ja202726y.

[34] Cortés-Ciriano I, Bender A. Concepts and applications of conformal prediction in computational drug discovery. *Wiley Interdisciplinary Reviews: Computational Molecular Science*, 9(5):e1410, 2019. doi:10.1002/wcms.1410.